

Protocol for efforts calculus on DasyLab 11.0

To best lead data treatment, follow the steps below:

1. The Software interface

-Launch DasyLab 11.0

-If you are treating information issued from trials done on AISI 304L, open the file: "Calculus_LD_304L.DSA"

-If you are treating information issued from trials done on 42 Cr Mo 4 QT, open the file: "Calculus_LD_42CD4.DSA"

-The following screen will appear:

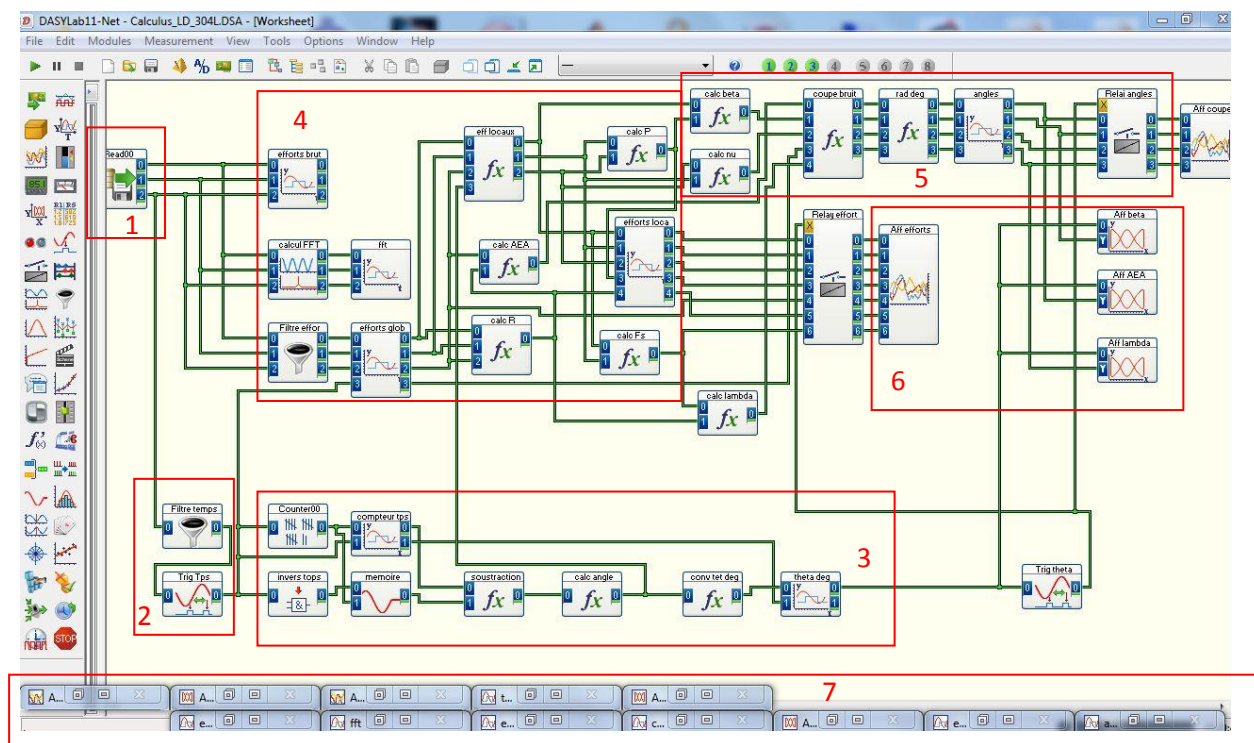


Fig.1. Calculus_LD_304L worksheet

- 1: Reading files module
- 2: Filter and trigger to define a time base
- 3: Modules to calculate and show attack angle θ
- 4: Modules to filter signals, calculate global and local efforts components, and show them
- 5: Modules to calculate angles and display them
- 6: Modules to show the associated curves
- 7: Curves views

2. The program preparation

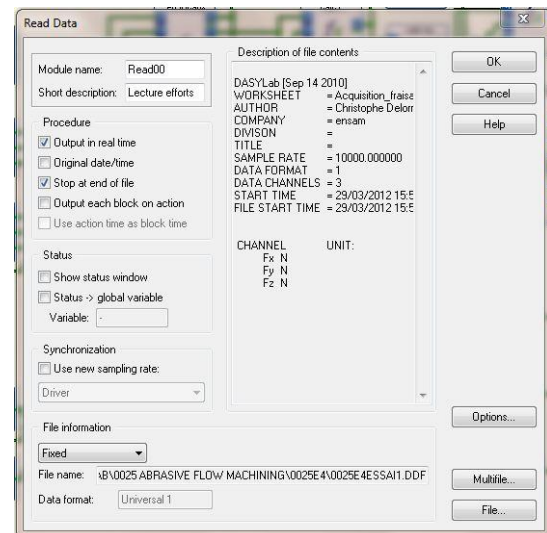
-Right click on the “Reading files” module (N°1). It is a module able to read data

-Click “File”

-Choose the file to be read

-Click “OK”

Now the file is ready to be simulated.



However, there are other parameters to be changed:

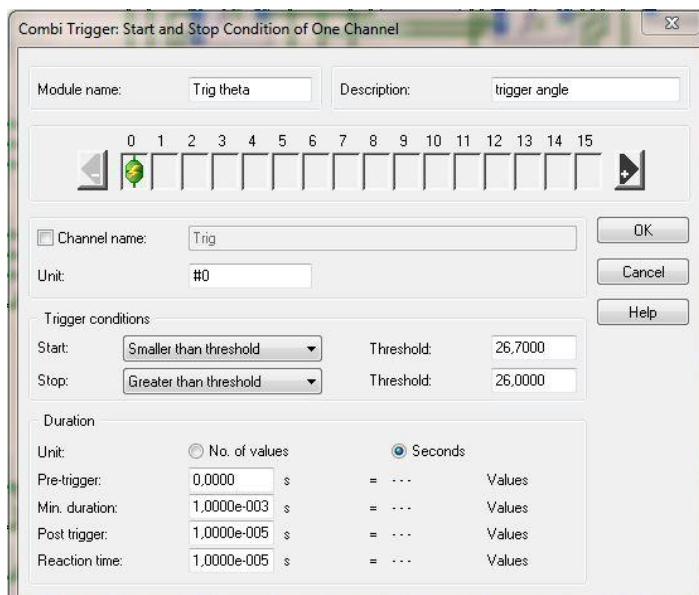
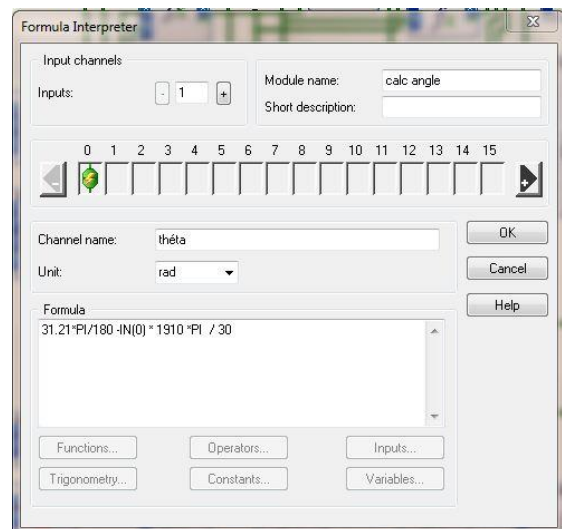
-Right click on the “Calc angle” module (in group N°3). It is a module able to interpret formula

-Click “Module properties”

-In the zone “Formula”, change the first value by the initial value of the attack angle expressed in degrees (in this figure, it is 31.21)

-Click “OK”

-Right click on the module called “Trig theta” (group N°3)



-In the zone “Threshold” write the value of the attack angle in degrees when the tooth is totally engaged. This module aims to get rid of the measurement in the transitory regim, when the cutting edge is partially engaged in the work material.

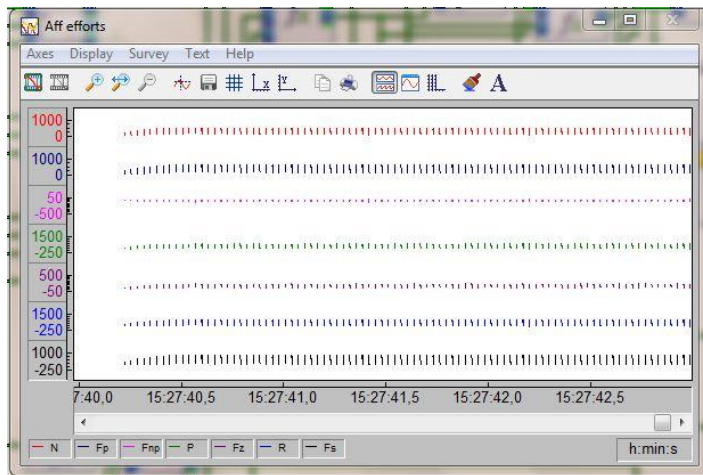
-Click “ok”

Now, our program is ready for simulation.

3. Simulation

-Click on the green triangle button





- Open the viewer called "Aff efforts" (see group N°7)

- Here you can see the evolution of the signals in real time.

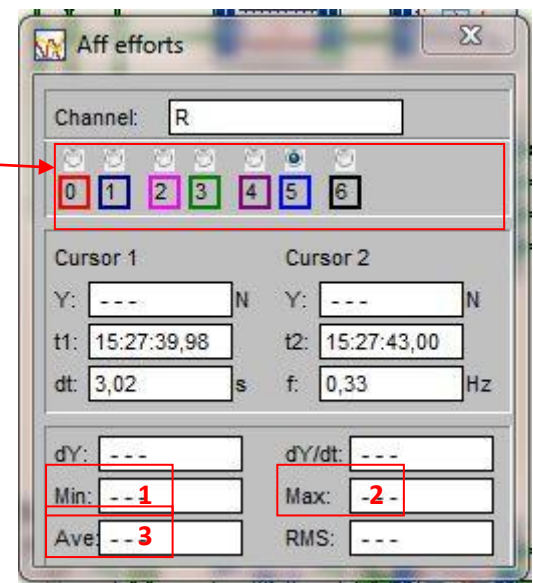
- When the permanent regime is established, click on pause button " || "



- Click on the button:

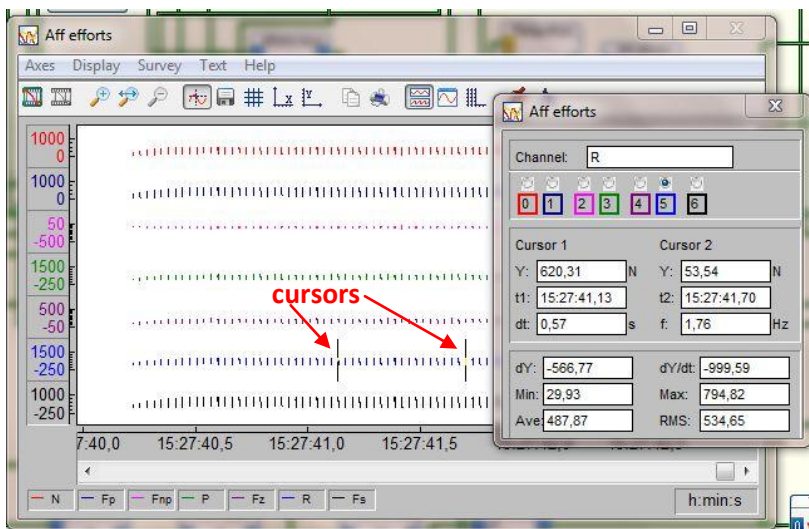


- A window like the shown on the right is then viewed
- Remark that there are two cursors on the curves.
- Choose the wanted signal to be measured from the listed buttons here
- The name of the corresponding effort component appears in the area called "Channel"
- In order to make a sampling of the measured signal, select an area by moving the two cursors



- In "Aff efforts" window, you can notice that there are values indicated.

- Pick up the wanted values: minimum value (1), maximum value (2), and average value (3)

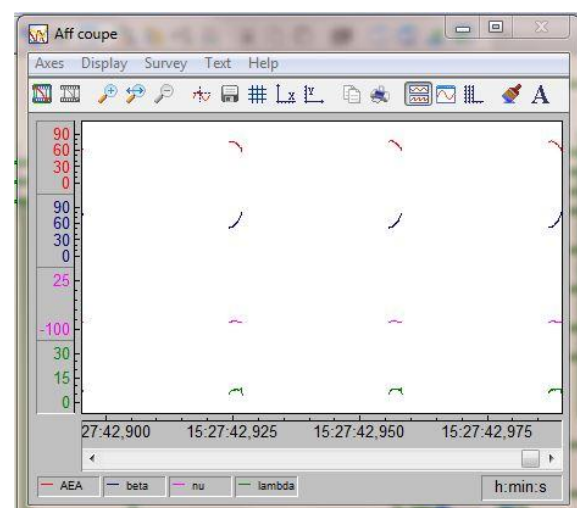


3. Measurements

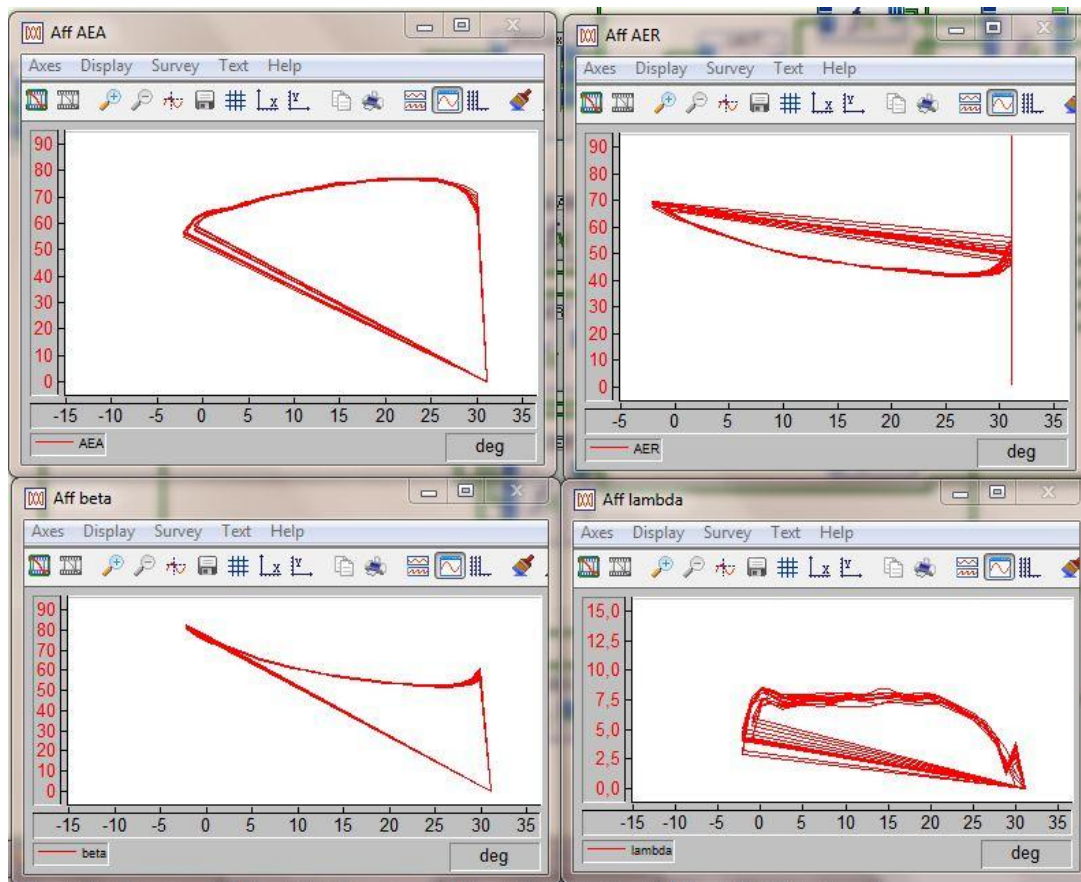
For both efforts components measurement and angles measurements, we do the same steps described previously. However, be sure of taking them from the appropriate windows:

- "Aff efforts" for efforts measurement

- "Aff coupe" for angles measurement



- You can also visualize the evolution of the local angles with the variation of theta in the following windows:



Windows showing the evolution of the local angles with that of attack during milling cycles

4. From a trial to another

The same steps are taken for each trial. To change from a trial to another, just change the read file name, and so repeat the same steps beginning with that of "Simulation".